Page-aware document evidence

ABSTRACT

This synthetic teaching paper studies page-aware evidence linking for local document readers.

The core question is whether stable source identifiers can preserve traceability through chunked analysis.

Our contribution is a source-first pipeline that keeps page numbers and verbatim text beside each finding.

The teaching example contains no real experiment and no external research claims.

INTRODUCTION

Long documents can hide important limitations near the end. Processing only the abstract loses this evidence.

We use a local PDF extractor followed by bounded model requests and a self-contained reading dashboard.

Method: sources before summaries

METHOD

Each paragraph receives a stable source identifier and the physical PDF page number.

The model returns a concise finding, a reader-friendly explanation, and a verbatim evidence quote.

An exact normalized text match checks that the quoted words exist in the extracted source.

A matching quote does not prove that a model conclusion is semantically correct.

ARGUMENT

First preserve the sources, then analyze each chunk, and finally merge findings without deleting leaves.

A controlled illustration

ILLUSTRATIVE EVALUATION

The fixture uses a deterministic local response server to test the complete software path. It is not an evaluation of a real language model and cannot establish scientific accuracy. The dashboard includes searchable nodes, evidence panels, diagram switching, and offline exports.

The implementation records analyzed chunk identifiers so omitted processing steps remain visible.

Limits and application boundary

LIMITATIONS

This demonstration uses synthetic text and does not measure performance on real scientific papers.

Image-only scans require OCR before use in the text-PDF baseline.

Evidence matching confirms text presence rather than entailment of the model claim.

APPLICATION BOUNDARY

The tool supports assisted reading, while numerical claims and scientific conclusions need human checking.

TAIL_SENTINEL: The last page must be analyzed and included in the coverage report.